



ABC Call Volume Trend Analysis

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PROJECT DESCRIPTION

ABC Call Volume Trend Analysis refers to the act of reviewing and analyzing the fluctuations, patterns, and trends in call volume specific to the "ABC" organization or system.

ABC company analysis might include:

- Identifying specific trends in call volume, such as peak hours (e.g., morning, afternoon), high-demand days (e.g., Mondays or weekends)
- Helping ABC optimize staffing and operational strategies based on expected call traffic, ensuring that the right number of agents are available during peak times without overstaffing during slower periods.

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Assumptions:

Working days: 6 days a week

Each Agent : 4 unplanned leaves per month;

An Agent's total working hours :9 hours

Break (lunch & snacks): 1.5 hours

Actual working hours: 7.5 hours

**Call working hours is 60% of actual working hours
:4.5 hours**

Total number of days in a month is 30.

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APPROACH

DATA UNDERSTANDING & CLEANING

Total Rows: 117989
Total Columns: 13
Null Values: 47877 Replaced
Them With Mode
Unusual Values: In
Agent_Name & Agent_ID
34198 Cells Replaced them
all with Abandoned Call cause
after looking at it thoroughly
found that these all were
referring with call status
were abandon too

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DATA ANALYSING

involves sorting data to
make it less complicated
while performing tasks and
By following this approach,
I was able to efficiently
analyze call volume trends
for ABC Company, identify
patterns, forecast future
volumes, and provide data-
driven recommendations to
optimize call center
operations.

DATA VISUALIZATION & INSIGHTS

involves Creating reports
and charts to
communicate findings
Extracting actionable
insights from the data
Creating documentation
so anyone going through
can understand the data
analysis process mostly
used PivotTables,
Line/column Charts

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TECH STACK USED

Microsoft Excel: To organize, analyze, and present data in a simple yet powerful manner

Microsoft Word: For documenting, editing, and sharing the process what I've done throughout the project

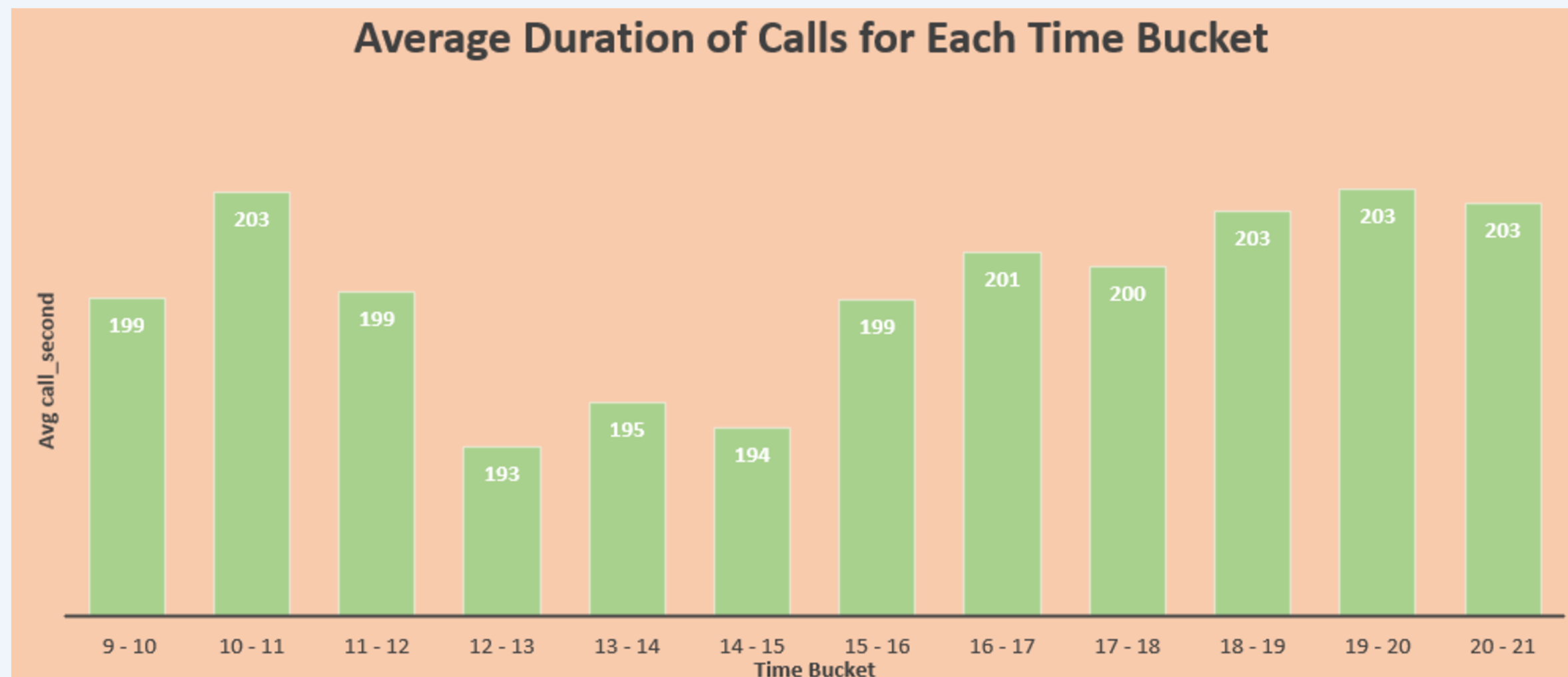
Canva & Microsoft power point: To use the proper templates and presenting project very appealing and interesting

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INSIGHTS

1. **Average Call Duration:** Determine the average duration of all incoming calls received by agents. This should be calculated for each time bucket.

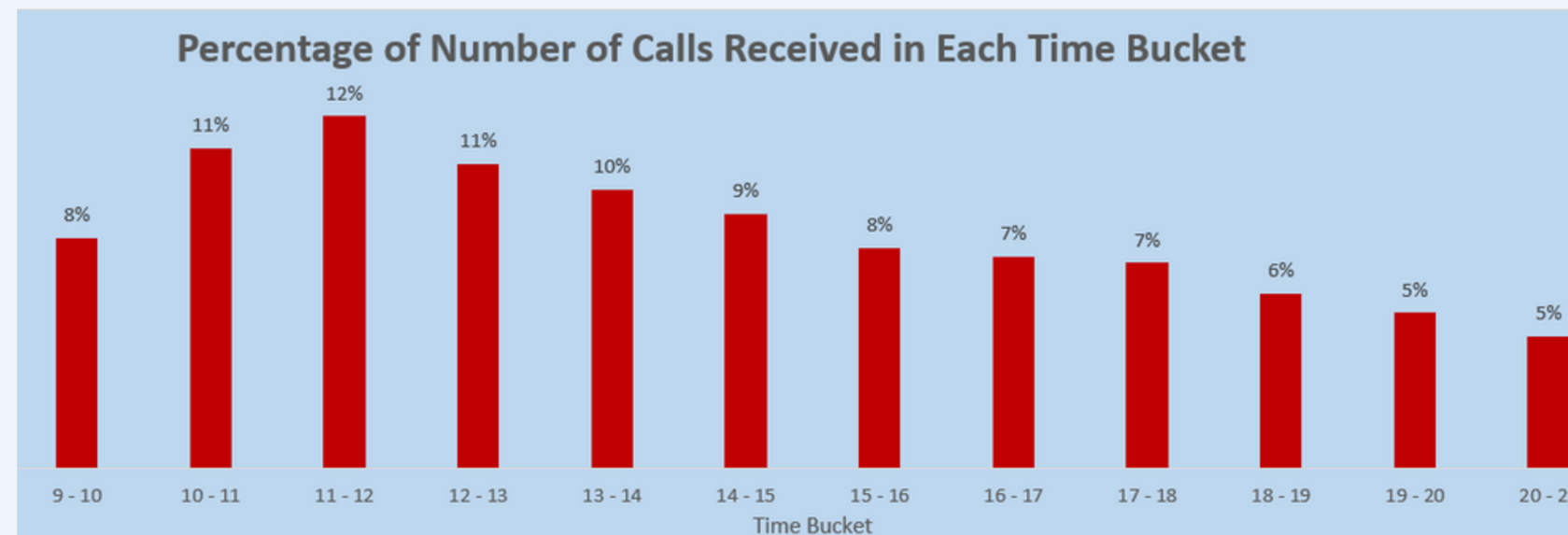
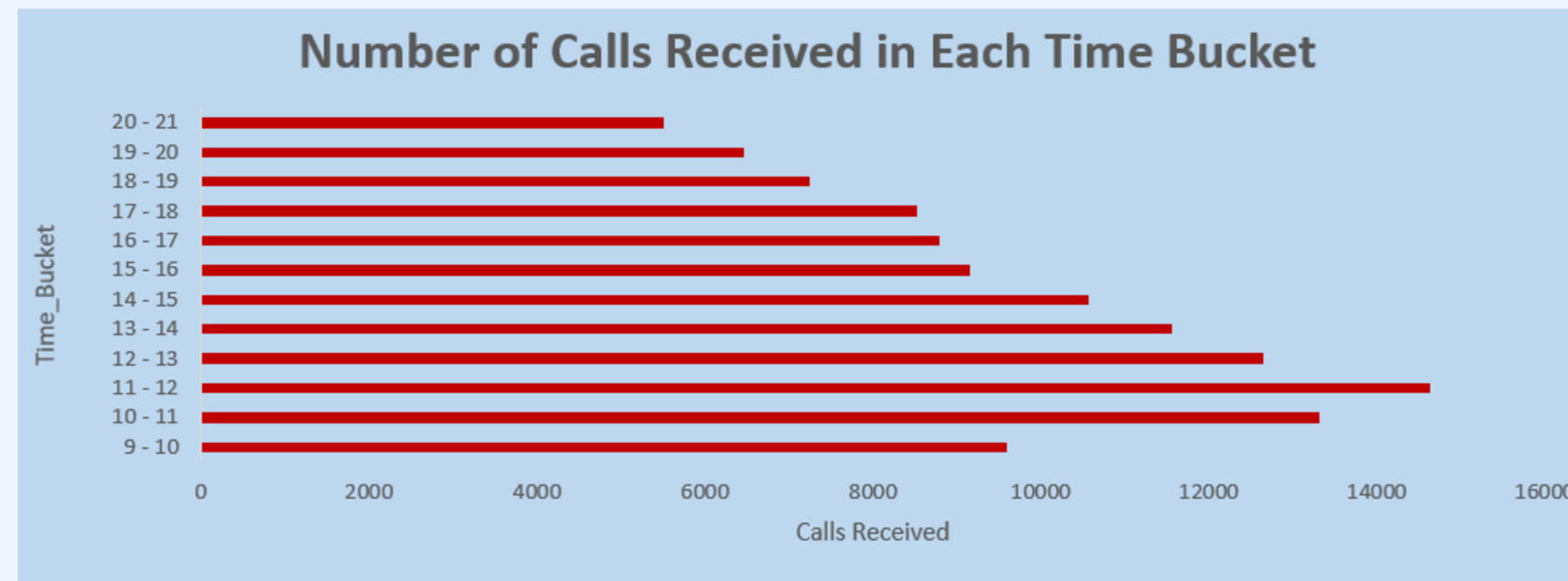


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INSIGHTS

2.Call Volume Analysis: Visualize the total number of calls received.
This should be represented as a graph or chart showing the number of calls against time. Time should be represented in buckets (e.g., 1-2, 2-3, etc.).

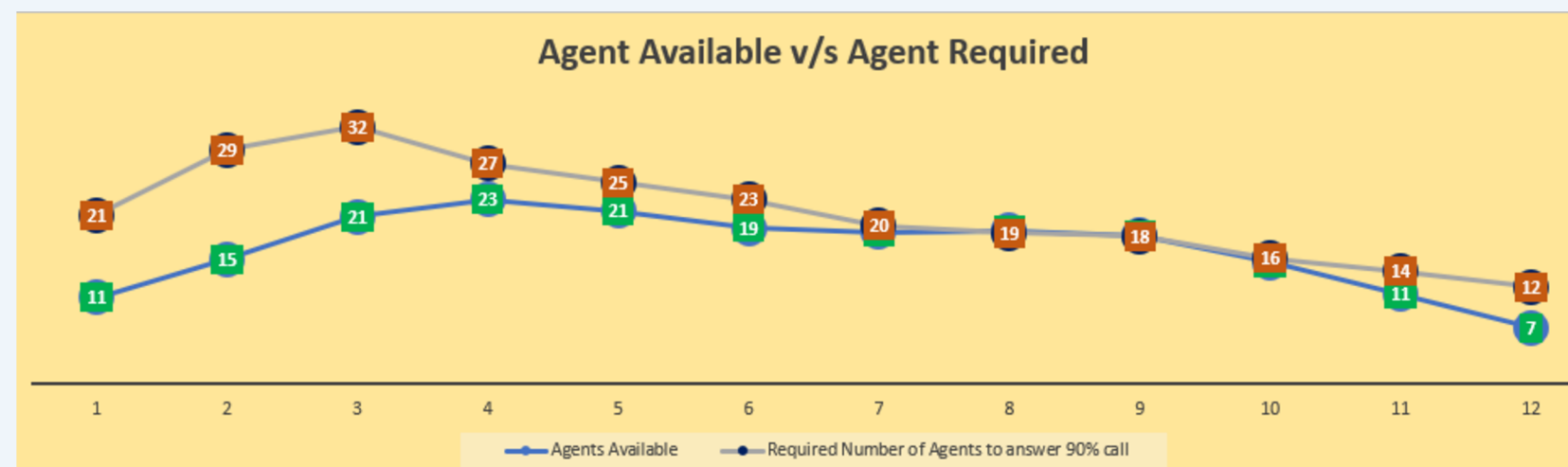
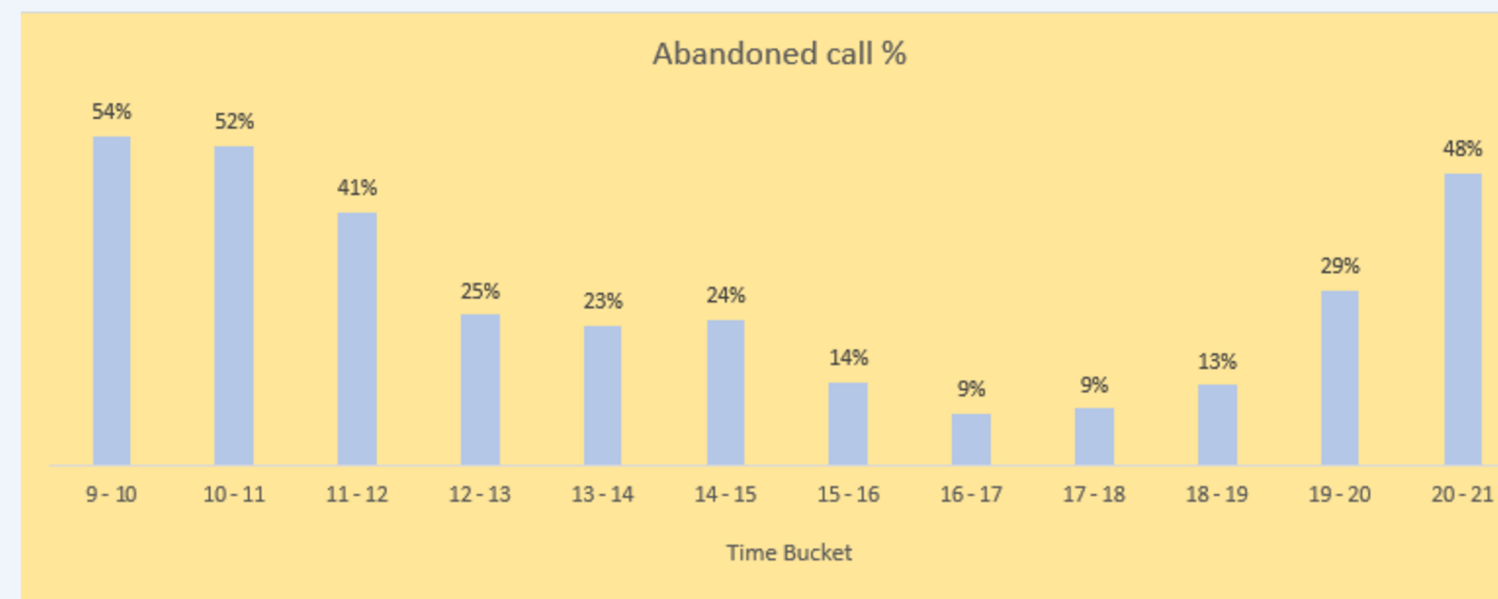


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INSIGHTS

3. Manpower Planning: The current rate of abandoned calls is approximately 30%. Propose a plan for manpower allocation during each time bucket (from 9 am to 9 pm) to reduce the abandon rate to 10%. In other words, you need to calculate the minimum number of agents required in each time bucket to ensure that at least 90 out of 100 calls are answered.



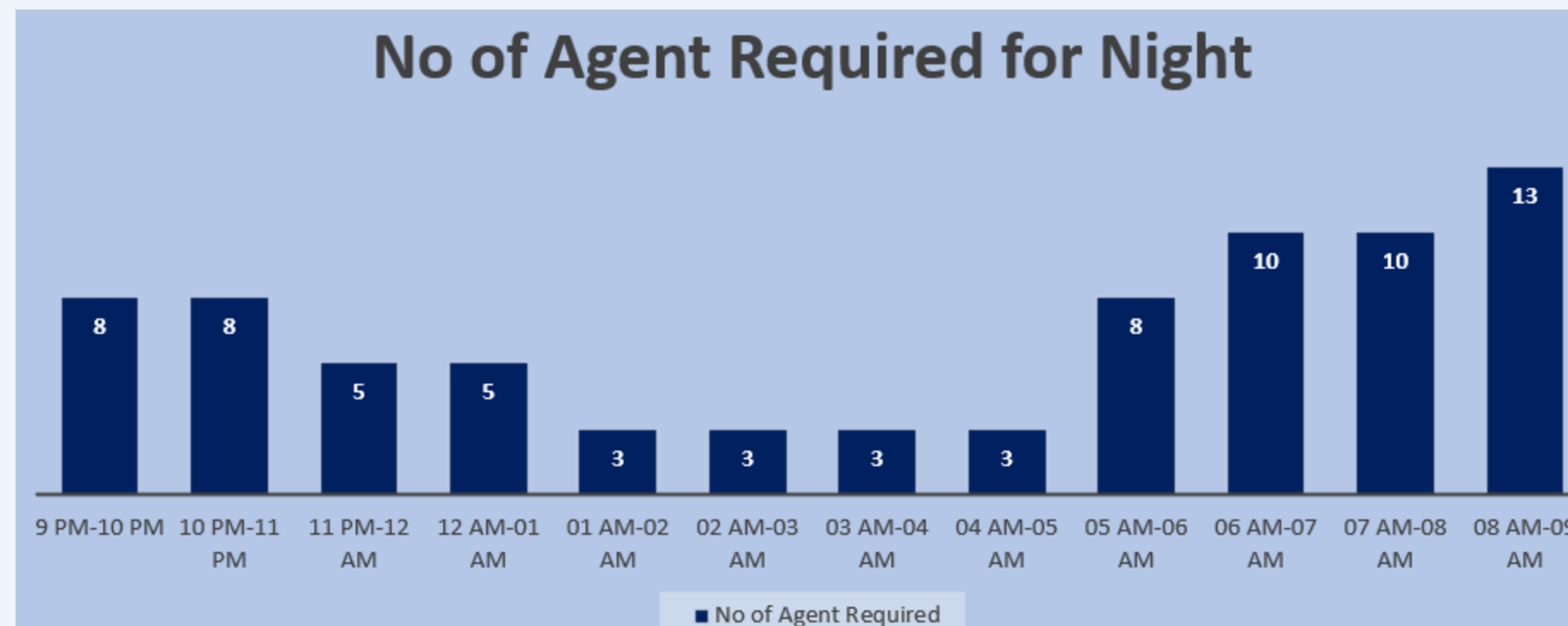
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INSIGHTS

4. Night Shift Manpower Planning: Customers also call ABC Insurance Company at night but don't get an answer because there are no agents available. This creates a poor customer experience. Assume that for every 100 calls that customers make between 9 am and 9 pm, they also make 30 calls at night between 9 pm and 9 am.

The distribution of these 30 calls is as follows:

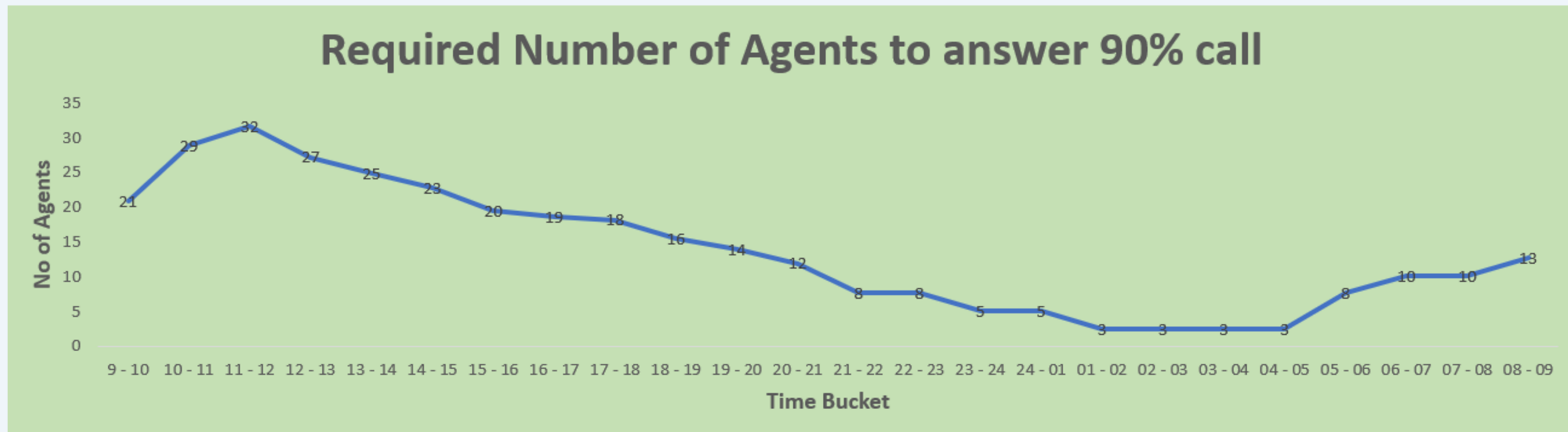


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CONCLUSION

Plan for Each Time Bucket Throughout the Day & Night



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THANK YOU

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